CASE STUDY

The operating principle of SSL using public key encryption could be easily understood with the following scenario "kumar orders a mobile phone from an online store (abc.com)."

1. Kumar connects to abc.com website through a secure connection, from his computer browser.
2. The abc.com website sends Kumar an digital certificate and a public key (P). This digital certificate issued by a certification authority (CA) proves the identity of abc.com.
3. Kumar's browser checks the certificate. It (browser) then agrees with the remote server on a symmetric cryptographic system to use. Then it randomly choose a key for this algorithm (session key K).
4. Kumar's browser sends P (K) to abc.com. Using its secret key S, the abc.com server calculates S (P (K)) = K. Thus, Kumar's browser and abc.com server are in possession of the same key.
5. Kumar enters his credit card number and other data. They constitute the "information". The browser sends these "information" to abc.com, encrypted using the key K. It also sends a summary of this "information", using a mathematical function called hash function.
6. With the K key, the abc.com server can dencrypt the "information". It also calculates the summary of information, and compares with the summary sent by Kumar's browser. If they coincide, it is assumed that the data has been correctly transmitted.

POINTS TO REMEMBER

- Phishing: Phishing is acquiring critical data like login credentials through telephone, sms, e-mail or any social media by the crackers disguising as authentic.
- Authentication: Information of the entity sending the document is often included in the document, but the information may be inaccurate. A digital signature can be used to authenticate the source of a document.
- Integrity: In many scenarios, the sender and receiver of the document will require confidence that the document has not been tampered with during the transfer. If the document was digitally signed, any modification of the document will invalidate the signature.
- Non-repudiation: Repudiation refers to any act of relinquishing responsibility for a message. Non-repudiation ensures that the signer who digitally signed the document cannot deny having signed it. The digitally signed documents strengthen

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